

Reprocessing of single use devices policy

Background

Single use devices (SUDs) are designed for immediate disposal after utilization on single patients. However, there is an increasing trend towards reprocessing SUDs due to increased cost of many sophisticated patient care devices especially in limited resource countries and when patients pay for their medical care.

There are risks associated with reprocessing of SUD because the manufacturer does not state the appropriate method for reprocessing. Therefore, the chosen reprocessing method can change the original specifications of the device rendering it unsafe for use due to the effect on physical integrity of the device causing risks as outlined below:

- Functionality and performance change/loss
- Mechanical injury
- Infection
- Toxicity

Therefore, regulations are required to control this process to prevent adverse events due to reuse of SUDs. Some countries such as US, Canada and Australia have strict legislations that regulate reprocessing of SUD by hospitals or by third party (other approved hospital or manufacturers).

In Oman there are no existing legislations to control this process. Hence, decision to reprocess SUD must be studied carefully and thoroughly using standardized and consistent procedure.

Definitions:

Label: A label is a display of written, printed, or graphic matter upon the immediate container of any device.

Single-use device (SUD): A single-use device, also referred to as a disposable device, intended for use on one patient during a single procedure. It is not intended to be reprocessed (cleaned, disinfected/sterilized) and used on another patient.

Reuse: The repeated use or multiple use of any medical device including devices intended for reuse or single-use, with reprocessing (cleaning, disinfection or sterilization) between uses.

Reprocessing: Reprocessing includes all the steps performed to make a contaminated reusable or single-use device ready for patient use. The steps may include cleaning, functional testing, repackaging, relabeling, disinfection or sterilization.

Reusable device: A reusable device is device that health care providers can reprocess and reuse on multiple patients.

Spaulding Classification: Spaulding classification for reusable instruments based on their risk of transmitting infection and according to the areas of the body into 3 classes:

- Critical devices, such as surgical forceps, come in contact with blood or normally sterile tissue.
- Semi-critical devices, such as endoscopes, come in contact with mucus membranes.
- Non-critical devices, such as stethoscopes, come in contact with unbroken skin.

CSSD: Central sterilization supply department is a specialized area responsible for the collection, decontamination, assembling, packing, sterilization, storing and distribution of sterile goods and equipment to patient care areas.

Scope

This policy applies to all hospital staff that utilizes medical devices in patient care, infection prevention and control, staff who are involved in reprocessing medical devices including but not limited to CSSD staff.

Policy statement:

Single use devices (SUD) are disposed of according to the manufacturer instruction for use after single use in single patients. SUD must not be reprocessed without approval of infection prevention and control (IPC). SUD can be considered for reprocessing only after thorough risk assessment and approval by IPC.

Roles and responsibilities:

All hospital staff:

- All staff must comply with the terms in this policy. Nonconformance with statements in this policy is considered a violation.
- Put a request for evaluation of SUD reuse to CSSD & IPC
- Maintain records that enables tracking of patients involved in reuse of SUD
- Record and report incidents of adverse events related to reuse of SUD in the incident notification system (Aman).

Infection prevention and control department:

- Develops this policy
- Monitors implementation of this policy
- Performs risk assessment in collaboration with other stakeholders
- Maintain and update list of reprocessed SUDs
- Review standard operating procedures for each approved SUD reuse
- Audit standards of reprocessing in CSSD

CSSD:

- Responsible for reprocessing of SUD once approved by IPC
- Write standard operating procedures for each approved SUD reuse
- Implement strict quality control measures for reprocessing
- Record and maintain registry of important details that enables tracking of devices

Quality and patient safety management:

- Reviews the terms of this policy
- Monitors adverse events related to this policy

Infection prevention and control committee:

Reviews and approves this policy.

Administration:

Provides governance on the implementation and violation of this policy

Steps for SUD processing approval:

- **Request for reprocessing SUD:**
Users should send a request in **AL Barwa** to CSSD for the need of reprocessing a SUD through Director of IPC utilizing the template in **Appendix 1**.
- **Risk assessment:**
After receiving the request, a risk assessment will be performed with the user to assess eligibility of the SUD for reuse and reprocessing utilizing the tool in **Appendix 2**.
Parameters to be assessed:
 - 1- Spaulding classification:
Risk is classified from low → high: non-critical → Semi-critical → Critical
 - 2- Intended use:

Use of SUD e.g. in visualization carries less risk than invasive interventional procedures

3- Features of device:

Smooth, simple, solid device has less risk than complex, hollow devices with multiple accessories

4- Reprocessing method

Compatibility of material to the reprocessing method decreases associated risk especially if the method used by manufacturer is the same chosen method for reuse.

○ **Approval for reprocessing:**

Decision to reject or approve the request will depend on the risk assessment and feasibility of reprocessing.

If the SUD is considered eligible, the following is required:

○ Method for reprocessing will be decided based on:

- Method used by manufacturer
- Spaulding class
- Material compatibility
- Temperature limit

○ Frequency:

Start with low frequency and can be reassessed after several uses.

○ **Processing:**

- An SOP will be written for the reprocessing of each SUD
- Pre-cleaning step by the user using compatible detergent
- Device sent to the CSSD covered in a container
- Cleaning step in the washer disinfectant
- Drying
- Mark the number of reprocessing cycles with water proof marker
- Package
- Labeling: name of the device, expiry date, caution to check integrity and functionality
- Registry: Name and hospital ID number, serial number of the device, number of reprocessing cycles.
- Quality control:
 - Cleaning: protein testing
 - Sterilization: Chemical indicator, biological indicators, cycle critical parameters
 - Maintenance records of the CSSD equipment

○ **Re-utilization:**

○ Registry: The following should be recorded by the user:

- Serial number
- Patient name and hospital number
- Operator
- Date
- Type of procedure
- What number of reprocessing cycles

- Check expiry date and details on the package and number of reprocessing cycles.
- Check physical integrity: color, texture, presence of holes or obvious damage to the device
- Check functionality before use on the patient without breaching sterility e.g. mobility
- Check performance of device for intended use
- Observe patient for adverse events
- Report adverse events in Aman system, IPC and CSSD
- Track other patients where the SUD has been used for.

Monitoring compliance:

- IPC to audit compliance to this audit biannually.
- Annual assessment of CSSD standards.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Reuse of Single-Use Devices: Understanding Risks and Strategies for Decision-Making for Health Care Organizations A White Paper by Joint Commission International	Joint Commission International	2017	
GCC Infection Prevention & Control Manual 3 rd edition	GCC center for infection control	2018	339-341
Reprocessing of Medical Products in Electrophysiology. Arq Bras Cardiol. 2017; 108(2):169-172	Ricardo Ryoshim Kuniyoshi et al.	2017	
Reprocessing of electrophysiology material in EHRA countries: an EHRA Young EP survey. Europace (2020) 00, 1–7 doi:10.1093/europace/euaa250	David Duncker.et al.	2020	
Labeling Recommendations for Single-Use Devices Reprocessed by Third Parties and Hospitals; Final Guidance for Industry and FDA	U.S. Department Of Health and Human Services Food and Drug Administration Center for Devices and Radiological Health	2001	

Appendices:

Appendix 1

Request for for reprocessing of Single use device

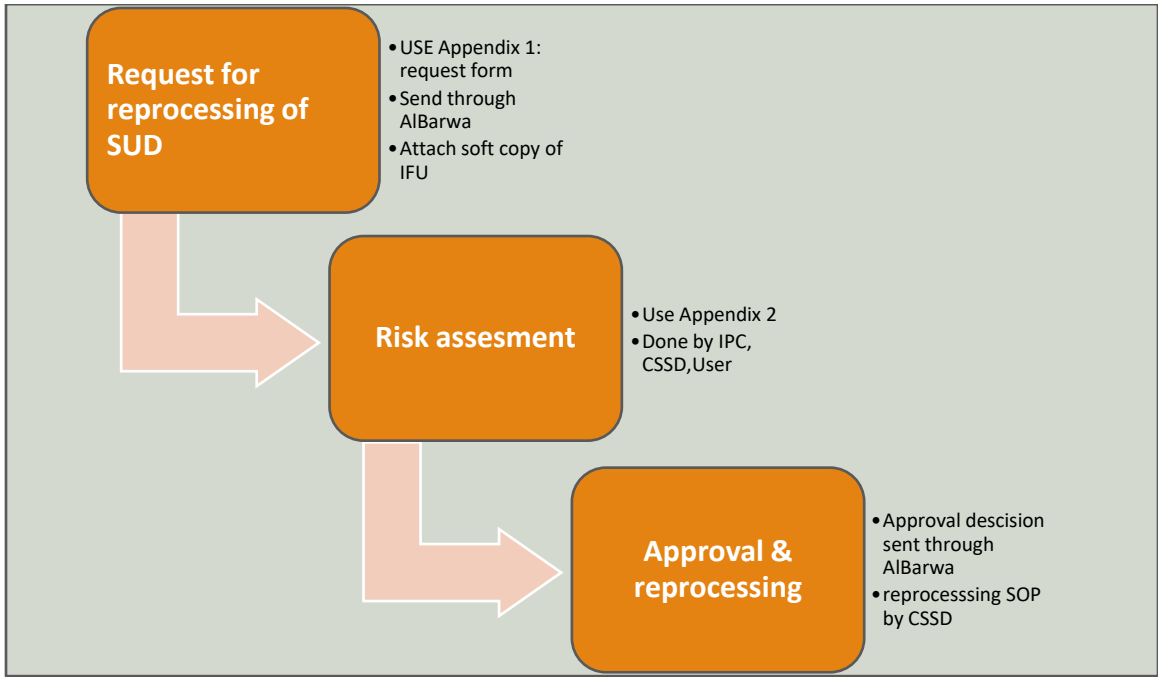
Department	Date
Requester's name	

Name of the device As stated in the IFU	
Intended use As stated in IFU	
The reason for reprocessing	1. 2.
Manufacturer instruction for use are attached Please attach a soft copy in AL Barwa and state yes or no.	Yes ☐ No ☐
State what is the composition material of the device Official evidence from supplier	

Appendix 2:
Risk assessment tool to assess eligibility of Single use device (SUD) for reprocessing

Assessors:		1.0	Date
		2.0	
		3.0	
Name of device		Owner	
1	Type of device List all components and numbers and their Spaulding classification and whether they can be disassembled	Non-critical:	1. 2.
		Semi-critical	1. 2.
		Critical	1. 2.
2	Intended use		
3	Features of device Describe the shape, complexity, attached accessories, hollow or solid		
4	Manufacturer label Single use Single use, don't sterilize Sterilization forbidden		
5	Sterilization method used by manufacturer		
6	Temperature limit Put T in °C&°F	Heat tolerant	Temp:
		Heat sensitive	Temp:
7	Material List all materials included in composition. (get details officially from supplier)		
8	Proposed Reprocessing method	High level disinfection	Method:
		Sterilization	Method:
9	Risks associated with reprocessing List all types of risks associated with the device	Functionality loss	
Infection			
Mechanical damage			
Toxicity			
Others			
10	How to mitigate risks		
11	Other special considerations		
12	Reasons for reprocessing request	1. 2.	

Appendix3: Algorithm for approval process of reprocessing of SUD



Review / Revision History

Revision History			
Revision	Description of change	Author	Date
1	Initial Release	Dr. Amal Al Tai	25/03/2024
2			
3			
4			
5			
Written by		Checked by	Authorized by
Amal AL Tai		Multi stakeholder team	IPC committee